

Conjecture 00000008228 is false:
the Ehrhart function of a rank-one weight polytope is never monic

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1 The conjecture

The English text of conjecture 00000008228 reads:

Definition: The extremities of weight vectors in geometric Satake: the vertices `sat_vertex` of the weight polytope under the Satake equivalence on the affine Grassmannian. Conjecture: `sat_vertex` (the vertices of the Satake polytope) are the weights of Mirkovic-Vilonen cycles (an MV-weight law); the volume of the polytope is the normalization of the representation dimension over $|W|$ (a volume-dimension normalization law); the count of extremal points (vertex count) equals the count of extremal elements of the saturated set of the weight lattice (an extremal count law); and the Ehrhart function of the polytope is a monic polynomial of degree the semisimple rank (a semisimple rank law).

The conjecture is a conjunction of four laws. We refute the fourth one, the *semisimple rank law*: “the Ehrhart function of the polytope is a monic polynomial of degree the semisimple rank”. Hence the conjunction is false. We do not use, and make no claim about, the first three laws (Remark 6 comments on the second one but is not part of the proof).

2 Reading and definitions

No group is fixed by the text, so the laws are asserted for a reductive group G and a coweight λ in general; one pair (G, λ) violating the fourth law refutes it. We take $G = SL_2$ and $G = PGL_2$.

Definition 1 (Root datum, Weyl group). A root datum is a pair of free abelian groups $X, X^\vee$ of finite rank in perfect pairing $\langle \cdot, \cdot \rangle$, with roots $\Phi \subset X$ and coroots $\Phi^\vee \subset X^\vee$, $\langle \alpha, \alpha^\vee \rangle = 2$, and both sets stable under the reflections $s_\alpha(x) = x - \langle x, \alpha^\vee \rangle \alpha$ (and dually). The Weyl group W is the group generated by the reflections. In Lean this is `Mathlib’s RootDatum` (a `RootPairing` over $\mathbb{Z}$) and `RootPairing.weylGroup`, the subgroup generated by the reflections.

For $a, b \in \mathbb{Z}$ with $ab = 2$, let $\mathcal{R}(a, b)$ be the root datum with $X = X^\vee = \mathbb{Z}$, pairing $\langle x, y \rangle = xy$, roots $\pm a$ and coroots $\pm b$ (Lean: `rk1 a b hab`). The datum $(a, b) = (2, 1)$ is that of SL_2 (`rootDatumSL2`) and $(a, b) = (1, 2)$ is that of PGL_2 (`rootDatumPGL2`). In geometric Satake the polytope lives in the cocharacter lattice of G , which is the character lattice of the dual group $G^\vee$. For $G = PGL_2$ this is the SL_2 datum, and for $G = SL_2$ it is the PGL_2 datum. All four solutions of $ab = 2$ are covered, so this choice does not matter.

Definition 2 (Weight polytope). For $\lambda \in X$, the weight polytope is $P_\lambda = \text{conv}(W \cdot \lambda) \subset X \otimes \mathbb{R}$. For $X = \mathbb{Z}$ we identify $X \otimes \mathbb{R} = \mathbb{R}$ (Lean: `weightPolytope P lam`, the real convex hull of the image of `MulAction.orbit P.weylGroup lam` under $\mathbb{Z} \rightarrow \mathbb{R}$).

Definition 3 (Ehrhart function). Following [1]: “Informally, if P is a polytope, and tP is the polytope formed by expanding P by a factor of t in each dimension, then $L(P, t)$ is the number of integer lattice points in tP .” For a lattice $\Lambda \leq X$ and $t \in \mathbb{Z}_{>0}$ we set $L_\Lambda(P, t) = \#\{x \in \Lambda : x \in tP\}$ (Lean: `ehrhart Lam Q t`, using `Set.ncard`; all sets counted here are finite).

Definition 4 (Semisimple rank). The semisimple rank is the rank of the root lattice $\mathbb{Z}\Phi \leq X$ (Lean: `semisimpleRank P`, the `Module.finrank` over $\mathbb{Z}$ of `P.rootSpan  $\mathbb{Z}$`). For $\mathcal{R}(a, b)$ it equals 1 (`semisimpleRank_rk1`).

Conventions. (i) Which lattice the Ehrhart function counts in is not fixed by the text. We allow every lattice $\Lambda \leq X$ that contains λ , which is exactly the condition that P_λ is a lattice polytope for Λ . This includes the full (co)weight lattice $\Lambda = X$, the root lattice $\Lambda = \mathbb{Z}\Phi$ when $\lambda \in \mathbb{Z}\Phi$, and the lattice $\mathbb{Z}\lambda$ generated by the vertices. (ii) If one counts in the coset $t\lambda + \mathbb{Z}\Phi$ (the weights of $V(t\lambda)$), this is the root-lattice count whenever $\lambda \in \mathbb{Z}\Phi$, so it is covered for such λ , for instance $\lambda = a$, the root. We do *not* cover this coset count for $\lambda \notin \mathbb{Z}\Phi$. Indeed, for the SL_2 datum and $\lambda = 1$ it gives $t + 1$, which *is* monic of degree 1. The law is still refuted under this reading, by $\lambda = a$. (iii) A polynomial “is” the Ehrhart function if it agrees with it at every positive integer t . We allow any real polynomial, which includes the rational ones. The hypothesis is only required for $t \geq 1$, so the statement is not weakened by the convention $L(P, 0) = 1$. (iv) We take $\lambda \neq 0$. For $\lambda = 0$ the polytope is a point and the case is degenerate.

3 The theorem

Theorem 5. *Let $ab = 2$, let $0 \neq \lambda \in X = \mathbb{Z}$, and let $\Lambda \leq \mathbb{Z}$ be a lattice with $\lambda \in \Lambda$. For $\mathcal{R}(a, b)$:*

1. *W acts on X as $\{\pm 1\}$, so $W \cdot \lambda = \{\lambda, -\lambda\}$ and $P_\lambda = [-|\lambda|, |\lambda|]$.*
2. *$L_X(P_\lambda, t) = 2|\lambda|t + 1$ for all t . More generally, if $\Lambda = d\mathbb{Z}$ then $L_\Lambda(P_\lambda, t) = 2ct + 1$ with $c = |\lambda/d| \geq 1$.*
3. *No real polynomial p with $p(t) = L_\Lambda(P_\lambda, t)$ for all integers $t \geq 1$ is monic, of any degree. In particular, none is monic of degree 1, the semisimple rank.*

Hence the semisimple rank law fails for SL_2 and for PGL_2 , for every nonzero λ and every lattice convention in (i). So the conjecture, being a conjunction, is false.

Proof. (1) For either root $\pm a$ with coroot $\pm b$ we have $s(x) = x - (x \cdot (\pm b))(\pm a) = x - abx = -x$. Every element of W is a product of reflections, so by induction on the generation of W it acts as $+1$ or -1 . The reflection lies in W , so $W \cdot \lambda = \{\lambda, -\lambda\}$. The convex hull of two points of $\mathbb{R}$ is the segment between them, which is $[-|\lambda|, |\lambda|]$.

(2) For $t \geq 0$ we have $tP_\lambda = [-t|\lambda|, t|\lambda|]$. Every subgroup of $\mathbb{Z}$ is $d\mathbb{Z}$ for some d (the ideal Λ of the PID $\mathbb{Z}$ is principal), and $\lambda \in \Lambda$ gives $\lambda = dm$ with $d, m \neq 0$. The map $k \mapsto dk$ is a bijection from $\{k : |k| \leq t|m|\}$ onto $\{x \in d\mathbb{Z} : |x| \leq t|dm|\}$, so $L_\Lambda(P_\lambda, t) = 2|m|t + 1$. For $d = 1$ this is $2|\lambda|t + 1$.

(3) If $p(t) = 2ct + 1$ for infinitely many t , then $p = 2cX + 1$, because two polynomials that agree at infinitely many points are equal. Its leading coefficient is $2c \geq 2$, which is not 1. $\square$

For example, for SL_2 and $\lambda = 1$ (the fundamental weight) in the weight lattice we get $L(t) = 2t + 1$. For $\lambda = a$ (the root) in the root lattice we get $L(t) = 2t + 1$ for both data. Two values already rule out a monic linear $t + c$, since it would force $L(2) - L(1) = 1$.

Remark 6 (Second law; not formalized, not used). For the SL_2 datum, $\text{vol}(P_n) = 2n$, while $\dim V(n)/|W| = (n+1)/2$. The ratio $4n/(n+1)$ is not constant in n , so no fixed normalization makes the second law true either. This remark is outside the Lean development, and the disproof does not depend on it.

Remark 7 (Scope). Only rank one is formalized. A computer check (in the attempt record, not in Lean) for A_2 with $\lambda = \omega_1$ in the weight lattice gives $L(1), \dots, L(4) = 4, 10, 19, 31$, whose leading coefficient is $3/2$.

4 Lean formalization

The file `lean/Conjecture8228/Basic.lean` (Lean 4, Mathlib v4.33.1, 268 lines) is built with `lake build`. Its main declarations are:

- `C8228.rk1 a b hab : RootDatum (Fin 2)  $\mathbb{Z}$   $\mathbb{Z}$ , with rootDatumSL2 = rk1 2 1, rootDatumPGL2 = rk1 1 2. The perfect pairing is LinearMap.mul.`
- `weyl_smul, orbit_eq`: W acts as ± 1 , and the orbit is $\{\lambda, -\lambda\}$.
- `weightPolytope_eq`: $P_\lambda = \text{Icc } (-|\lambda|) \ |\lambda|$.
- `semisimpleRank_rk1`: the semisimple rank is 1.
- `ehrhart_top`: $L_X(P_\lambda, t) = 2|\lambda|t + 1$ for every λ and t .
- `ehrhart_formula`: for $\lambda \neq 0$ and any lattice $\Lambda \ni \lambda$, $L_\Lambda(P_\lambda, t) = 2ct + 1$ with $c \geq 1$.
- `ehrhart_not_monico (main)`: for `lam  $\neq 0$` and `lam  $\in \text{Lam}$` ,
 $\neg \exists p : \mathbb{R}[X], p.\text{Monic} \wedge \forall t : \mathbb{N}, 0 < t \rightarrow p.\text{eval } t = \text{ehrhart Lam } (\text{weightPolytope } (\text{rk1 a b hab}) \text{ lam}) t$.
- `semisimple_rank_law_fails`: the same statement with the extra conjunct `p.natDegree = semisimpleRank (rk1 a b hab)`. Its specialization to the root lattice `(rk1 a b hab).rootSpan  $\mathbb{Z}$` is `semisimple_rank_law_fails_rootLattice`.

Axiom audit. `#print axioms` reports `[propext, Classical.choice, Quot.sound]` for `ehrhart_not_monico`, `semisimple_rank_law_fails`, `semisimple_rank_law_fails_rootLattice`, `ehrhart_top` and `semisimpleRank_rk1`. There is no `sorry` and no `native_decide`, and no axioms are added.

References

- [1] Wikipedia, *Ehrhart polynomial*, https://en.wikipedia.org/wiki/Ehrhart_polynomial (retrieved 2026-10-05).